

You need to deploy a new service to production. The service needs to automatically scale using a managed instance group and should be deployed across multiple regions. The service needs a large number of resources for each instance and you need to plan for capacity. What should you do?

- A. Monitor results of Cloud Trace to determine the optimal sizing.
 - B. Use the n2-highcpu-96 machine type in the configuration of the managed instance group.
 - C. Deploy the service in multiple regions and use an internal load balancer to route traffic.
 - D. Validate that the resource requirements are within the available project quota limits of each region. **Most Voted**
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Your organization stores all application logs from multiple Google Cloud projects in a central Cloud Logging project. Your security team wants to enforce a rule that each project team can only view their respective logs and only the operations team can view all the logs. You need to design a solution that meets the security team's requirements while minimizing costs. What should you do?

- A. Grant each project team access to the project _Default view in the central logging project. Grant logging viewer access to the operations team in the central logging project.
 - B. Create Identity and Access Management (IAM) roles for each project team and restrict access to the _Default log view in their individual Google Cloud project. Grant viewer access to the operations team in the central logging project.
 - C. Create log views for each project team and only show each project team their application logs. Grant the operations team access to the _AllLogs view in the central logging project. **Most Voted**
 - D. Export logs to BigQuery tables for each project team. Grant project teams access to their tables. Grant logs writer access to the operations team in the central logging project.
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Your company uses Jenkins running on Google Cloud VM instances for CI/CD. You need to extend the functionality to use infrastructure as code automation by using Terraform. You must ensure that the Terraform Jenkins instance is authorized to create Google Cloud resources. You want to follow Google-recommended practices. What should you do?

- A. Confirm that the Jenkins VM instance has an attached service account with the appropriate Identity and Access Management (IAM) permissions. **Most Voted**

- B. Use the Terraform module so that Secret Manager can retrieve credentials.
 - C. Create a dedicated service account for the Terraform instance. Download and copy the secret key value to the GOOGLE_CREDENTIALS environment variable on the Jenkins server.
 - D. Add the gcloud auth application-default login command as a step in Jenkins before running the Terraform commands.
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You encounter a large number of outages in the production systems you support. You receive alerts for all the outages, the alerts are due to unhealthy systems that are automatically restarted within a minute. You want to set up a process that would prevent staff burnout while following Site Reliability Engineering (SRE) practices. What should you do?

- A. Eliminate alerts that are not actionable **Most Voted**
 - B. Redefine the related SLO so that the error budget is not exhausted
 - C. Distribute the alerts to engineers in different time zones
 - D. Create an incident report for each of the alerts
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You are developing reusable infrastructure as code modules. Each module contains integration tests that launch the module in a test project. You are using GitHub for source control. You need to continuously test your feature branch and ensure that all code is tested before changes are accepted. You need to implement a solution to automate the integration tests. What should you do?

- A. Use a Jenkins server for CI/CD pipelines. Periodically run all tests in the feature branch.
 - B. Ask the pull request reviewers to run the integration tests before approving the code.
 - C. Use Cloud Build to run the tests. Trigger all tests to run after a pull request is merged.
 - D. Use Cloud Build to run tests in a specific folder. Trigger Cloud Build for every GitHub pull request. **Most Voted**
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Your company runs services by using Google Kubernetes Engine (GKE). The GKE clusters in the development environment run applications with verbose logging enabled. Developers view logs by using the `kubectl logs` command and do not use Cloud Logging. Applications do not have a uniform logging structure defined. You need to minimize the costs associated with application logging while still collecting GKE operational logs. What should you do?

- A. Run the `gcloud container clusters update --logging=SYSTEM` command for the development cluster.
 - B. Run the `gcloud container clusters update --logging=WORKLOAD` command for the development cluster.
 - C. Run the `gcloud logging sinks update _Default --disabled` command in the project associated with the development environment.
 - D. Add the severity `>= DEBUG` resource.type = "k8s_container" exclusion filter to the `_Default` logging sink in the project associated with the development environment. **Most Voted**
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You have deployed a fleet of Compute Engine instances in Google Cloud. You need to ensure that monitoring metrics and logs for the instances are visible in Cloud Logging and Cloud Monitoring by your company's operations and cyber security teams. You need to grant the required roles for the Compute Engine service account by using Identity and Access Management (IAM) while following the principle of least privilege. What should you do?

- A. Grant the `logging.logWriter` and `monitoring.metricWriter` roles to the Compute Engine service accounts. **Most Voted**
 - B. Grant the `logging.admin` and `monitoring.editor` roles to the Compute Engine service accounts.
 - C. Grant the `logging.editor` and `monitoring.metricWriter` roles to the Compute Engine service accounts.
 - D. Grant the `logging.logWriter` and `monitoring.editor` roles to the Compute Engine service accounts.
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You are the Site Reliability Engineer responsible for managing your company's data services and products. You regularly navigate operational challenges, such as unpredictable data volume and high cost, with your company's data ingestion processes. You recently learned that a new data ingestion product will be developed in Google Cloud. You need to collaborate with the product development team to provide operational input on the new product. What should you do?

- A. Deploy the prototype product in a test environment, run a load test, and share the results with the product development team.
 - B. When the initial product version passes the quality assurance phase and compliance assessments, deploy the product to a staging environment. Share error logs and performance metrics with the product development team.
 - C. When the new product is used by at least one internal customer in production, share error logs and monitoring metrics with the product development team.
 - D. Review the design of the product with the product development team to provide feedback early in the design phase. **Most Voted**
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You are investigating issues in your production application that runs on Google Kubernetes Engine (GKE). You determined that the source of the issue is a recently updated container image, although the exact change in code was not identified. The deployment is currently pointing to the latest tag. You need to update your cluster to run a version of the container that functions as intended. What should you do?

- A. Create a new tag called stable that points to the previously working container, and change the deployment to point to the new tag.
 - B. Alter the deployment to point to the sha256 digest of the previously working container. **Most Voted**
 - C. Build a new container from a previous Git tag, and do a rolling update on the deployment to the new container.
 - D. Apply the latest tag to the previous container image, and do a rolling update on the deployment.
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You need to create a Cloud Monitoring SLO for a service that will be published soon. You want to verify that requests to the service will be addressed in fewer than 300 ms at least 90% of the time per calendar month. You need to identify the metric and evaluation method to use. What should you do?

- A. Select a latency metric for a request-based method of evaluation. **Most Voted**
 - B. Select a latency metric for a window-based method of evaluation.
 - C. Select an availability metric for a request-based method of evaluation.
 - D. Select an availability metric for a window-based method of evaluation.
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You have an application that runs on Cloud Run. You want to use live production traffic to test a new version of the application, while you let the quality assurance team perform manual testing. You want to limit the potential impact of any issues while testing the new version, and you must be able to roll back to a previous version of the application if needed. How should you deploy the new version? (Choose two.)

- A. Deploy the application as a new Cloud Run service.
 - B. Deploy a new Cloud Run revision with a tag and use the --no-traffic option. **Most Voted**
 - C. Deploy a new Cloud Run revision without a tag and use the --no-traffic option.
 - D. Deploy the new application version and use the --no-traffic option. Route production traffic to the revision's URL. **Most Voted**
 - E. Deploy the new application version, and split traffic to the new version.
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You need to introduce postmortems into your organization. You want to ensure that the postmortem process is well received. What should you do? (Choose two.)

- A. Encourage new employees to conduct postmortems to team through practice.
 - B. Create a designated team that is responsible for conducting all postmortems.
 - C. Encourage your senior leadership to acknowledge and participate in postmortems. **Most Voted**
 - D. Ensure that writing effective postmortems is a rewarded and celebrated practice. **Most Voted**
 - E. Provide your organization with a forum to critique previous postmortems.
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You are the Operations Lead for an ongoing incident with one of your services. The service usually runs at around 70% capacity. You notice that one node is returning 5xx errors for all requests. There has also been a noticeable increase in support cases from customers. You need to remove the offending node from the load balancer pool so that you can isolate and investigate the node. You want to follow Google-recommended practices to manage the incident and reduce the impact on users. What should you do?

- A. 1. Communicate your intent to the incident team.
2. Perform a load analysis to determine if the remaining nodes can handle the increase in traffic offloaded from the removed node, and scale appropriately.
3. When any new nodes report healthy, drain traffic from the unhealthy node, and remove the unhealthy node from service. **Most Voted**
 - B. 1. Communicate your intent to the incident team.
2. Add a new node to the pool, and wait for the new node to report as healthy.
3. When traffic is being served on the new node, drain traffic from the unhealthy node, and remove the old node from service.
 - C. 1. Drain traffic from the unhealthy node and remove the node from service.
2. Monitor traffic to ensure that the error is resolved and that the other nodes in the pool are handling the traffic appropriately.
3. Scale the pool as necessary to handle the new load.
4. Communicate your actions to the incident team.
 - D. 1. Drain traffic from the unhealthy node and remove the old node from service.
2. Add a new node to the pool, wait for the new node to report as healthy, and then serve traffic to the new node.
3. Monitor traffic to ensure that the pool is healthy and is handling traffic appropriately.
4. Communicate your actions to the incident team.
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You are configuring your CI/CD pipeline natively on Google Cloud. You want builds in a pre-production Google Kubernetes Engine (GKE) environment to be automatically load-tested before being promoted to the production GKE environment. You need to ensure that only builds that have passed this test are deployed to production. You want to follow Google-recommended practices. How should you configure this pipeline with Binary Authorization?

- A. Create an attestation for the builds that pass the load test by requiring the lead quality assurance engineer to sign the attestation by using their personal private key.
- B. Create an attestation for the builds that pass the load test by using a private key stored in Cloud Key Management Service (Cloud KMS) with a service account JSON key stored as a Kubernetes Secret.

- C. Create an attestation for the builds that pass the load test by using a private key stored in Cloud Key Management Service (Cloud KMS) authenticated through Workload Identity. **Most Voted**
 - D. Create an attestation for the builds that pass the load test by requiring the lead quality assurance engineer to sign the attestation by using a key stored in Cloud Key Management Service (Cloud KMS).
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You are deploying an application to Cloud Run. The application requires a password to start. Your organization requires that all passwords are rotated every 24 hours, and your application must have the latest password. You need to deploy the application with no downtime. What should you do?

- A. Store the password in Secret Manager and send the secret to the application by using environment variables.
 - B. Store the password in Secret Manager and mount the secret as a volume within the application. **Most Voted**
 - C. Use Cloud Build to add your password into the application container at build time. Ensure that Artifact Registry is secured from public access.
 - D. Store the password directly in the code. Use Cloud Build to rebuild and deploy the application each time the password changes.
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Your company runs applications in Google Kubernetes Engine (GKE) that are deployed following a GitOps methodology. Application developers frequently create cloud resources to support their applications. You want to give developers the ability to manage infrastructure as code, while ensuring that you follow Google-recommended practices. You need to ensure that infrastructure as code reconciles periodically to avoid configuration drift. What should you do?

- A. Install and configure Config Connector in Google Kubernetes Engine (GKE). **Most Voted**
- B. Configure Cloud Build with a Terraform builder to execute terraform plan and terraform apply commands.
- C. Create a Pod resource with a Terraform docker image to execute terraform plan and terraform apply commands.
- D. Create a Job resource with a Terraform docker image to execute terraform plan and terraform apply commands.

You are designing a system with three different environments: development, quality assurance (QA), and production. Each environment will be deployed with Terraform and has a Google Kubernetes Engine (GKE) cluster created so that application teams can deploy their applications. Anthos Config Management will be used and templated to deploy infrastructure level resources in each GKE cluster. All users (for example, infrastructure operators and application owners) will use GitOps. How should you structure your source control repositories for both Infrastructure as Code (IaC) and application code?

- A. • Cloud Infrastructure (Terraform) repository is shared: different directories are different environments
 - GKE Infrastructure (Anthos Config Management Kustomize manifests) repository is shared: different overlay directories are different environments
 - Application (app source code) repositories are separated: different branches are different features **Most Voted**
 - B. • Cloud Infrastructure (Terraform) repository is shared: different directories are different environments
 - GKE Infrastructure (Anthos Config Management Kustomize manifests) repositories are separated: different branches are different environments
 - Application (app source code) repositories are separated: different branches are different features
 - C. • Cloud Infrastructure (Terraform) repository is shared: different branches are different environments
 - GKE Infrastructure (Anthos Config Management Kustomize manifests) repository is shared: different overlay directories are different environments
 - Application (app source code) repository is shared: different directories are different features
 - D. • Cloud Infrastructure (Terraform) repositories are separated: different branches are different environments
 - GKE Infrastructure (Anthos Config Management Kustomize manifests) repositories are separated: different overlay directories are different environments
 - Application (app source code) repositories are separated: different branches are different
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As a Site Reliability Engineer, you support an application written in Go that runs on Google Kubernetes Engine (GKE) in production. After releasing a new version of the application, you notice the application runs for about 15 minutes and then restarts. You decide to add Cloud Profiler to your application and now notice that the heap usage grows constantly until the application restarts. What should you do?

- A. Increase the CPU limit in the application deployment.
 - B. Add high memory compute nodes to the cluster.
 - C. Increase the memory limit in the application deployment. **Most Voted**
 - D. Add Cloud Trace to the application, and redeploy.
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You are deploying a Cloud Build job that deploys Terraform code when a Git branch is updated. While testing, you noticed that the job fails. You see the following error in the build logs:

Initializing the backend...

Error: Failed to get existing workspaces: querying Cloud Storage failed: googleapi: Error 403

You need to resolve the issue by following Google-recommended practices. What should you do?

- A. Change the Terraform code to use local state.
 - B. Create a storage bucket with the name specified in the Terraform configuration.
 - C. Grant the roles/owner Identity and Access Management (IAM) role to the Cloud Build service account on the project.
 - D. Grant the roles/storage.objectAdmin Identity and Access Management (IAM) role to the Cloud Build service account on the state file bucket. **Most Voted**
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Your company runs applications in Google Kubernetes Engine (GKE). Several applications rely on ephemeral volumes. You noticed some applications were unstable due to the DiskPressure node condition on the worker nodes. You need to identify which Pods are causing the issue, but you do not have execute access to workloads and nodes. What should you do?

- A. Check the node/ephemeral_storage/used_bytes metric by using Metrics Explorer.
 - B. Check the container/ephemeral_storage/used_bytes metric by using Metrics Explorer. **Most Voted**
 - C. Locate all the Pods with emptyDir volumes. Use the df -h command to measure volume disk usage.
 - D. Locate all the Pods with emptyDir volumes. Use the df -sh * command to measure volume disk usage.
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You are designing a deployment technique for your applications on Google Cloud. As part of your deployment planning, you want to use live traffic to gather performance metrics for new versions of your applications. You need to test against the full production load before your applications are launched. What should you do?

- A. Use A/B testing with blue/green deployment.
 - B. Use canary testing with continuous deployment.
 - C. Use canary testing with rolling updates deployment.
 - D. Use shadow testing with continuous deployment. **Most Voted**
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Your Cloud Run application writes unstructured logs as text strings to Cloud Logging. You want to convert the unstructured logs to JSON-based structured logs. What should you do?

- A. Modify the application to use Cloud Logging software development kit (SDK), and send log entries with a jsonPayload field.
 - B. Install a Fluent Bit sidecar container, and use a JSON parser.
 - C. Install the log agent in the Cloud Run container image, and use the log agent to forward logs to Cloud Logging.
 - D. Configure the log agent to convert log text payload to JSON payload. **Most Voted**
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You are leading a DevOps project for your organization. The DevOps team is responsible for managing the service infrastructure and being on-call for incidents. The Software Development team is responsible for writing, submitting, and reviewing code. Neither team has any published SLOs. You want to design a new joint-ownership model for a service between the DevOps team and the Software Development team. Which responsibilities should be assigned to each team in the new joint-ownership model?

- A.

DevOps team responsibilities	Software Development team responsibilities
• Manage the service infrastructure • Be on-call for incidents • Perform code reviews	• Submit code to be reviewed by the DevOps team • Publish the SLOs that the DevOps team must meet

- B.

DevOps team responsibilities	Software Development team responsibilities
• Manage the service infrastructure • Perform code reviews	• Submit code to be reviewed by the DevOps team • Be on-call for incidents • Publish the SLOs that the DevOps team must meet

- C. **Most Voted**

DevOps team responsibilities	Shared responsibilities	Software Development team responsibilities
• Manage the service infrastructure	• Perform code reviews • Be on-call for incidents on a rotation basis • Adopt and publish SLOs for the service	• Submit code to be reviewed

- D.

DevOps team responsibilities	Shared responsibilities	Software Development team responsibilities
• Manage the service infrastructure	• Perform code reviews • Be on-call for incidents on a rotation basis • Adopt and publish SLOs for the service	• Submit code to be reviewed

You recently migrated an ecommerce application to Google Cloud. You now need to prepare the application for the upcoming peak traffic season. You want to follow Google-recommended practices. What should you do first to prepare for the busy season?

- A. Migrate the application to Cloud Run, and use autoscaling.
- B. Create a Terraform configuration for the application's underlying infrastructure to quickly deploy to additional regions.
- C. Load test the application to profile its performance for scaling. **Most Voted**
- D. Pre-provision the additional compute power that was used last season, and expect growth.

You are monitoring a service that uses n2-standard-2 Compute Engine instances that serve large files. Users have reported that downloads are slow. Your Cloud Monitoring dashboard shows that your VMs are running at peak network throughput. You want to improve the network throughput performance. What should you do?

- A. Add additional network interface controllers (NICs) to your VMs.
- B. Deploy a Cloud NAT gateway and attach the gateway to the subnet of the VMs.
- C. Change the machine type for your VMs to n2-standard-8. **Most Voted**
- D. Deploy the Ops Agent to export additional monitoring metrics.

Your organization is starting to containerize with Google Cloud. You need a fully managed storage solution for container images and Helm charts. You need to identify a storage solution that has native integration into existing Google Cloud services, including Google Kubernetes Engine (GKE), Cloud Run, VPC Service Controls, and Identity and Access Management (IAM). What should you do?

- A. Use Docker to configure a Cloud Storage driver pointed at the bucket owned by your organization.
 - B. Configure an open source container registry server to run in GKE with a restrictive role-based access control (RBAC) configuration.
 - C. Configure Artifact Registry as an OCI-based container registry for both Helm charts and container images. **Most Voted**
 - D. Configure Container Registry as an OCI-based container registry for container images.
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